

Ensemble Strategies for Multilingual Polarization Detection: A Comprehensive Framework for SemEval–2026 Task 9

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Abstract

This paper presents a comprehensive framework for multilingual polarization detection developed for SemEval–2026 Task 9. We address three subtasks across English and Swahili: binary polarization detection (Subtask 1), multi-label polarization type classification (Subtask 2), and multi-label manifestation identification (Subtask 3). Our approach integrates transformer architectures with ensemble learning strategies, weak supervision via keyword systems, and class-aware loss functions to handle severe data imbalance and limited annotated resources for Swahili. The system, implemented in PyTorch and publicly available at [GitHub](#), achieves competitive results on the official Codabench evaluation: average Macro-F1 of 0.7803 for Subtask 1, 0.151 for Subtask 2, and 0.310 for Subtask 3. Notably, the system demonstrates strong Swahili performance in Subtask 1 (Macro-F1 = 0.8195), indicating effective cross-lingual transfer. We provide both theoretical analysis of ensemble variance reduction and practical insights into system implementation, offering clear pathways for improving multilingual content moderation systems.

1 Introduction

The detection of polarized and divisive content on social media platforms represents a critical challenge at the intersection of computational linguistics, political science, and social psychology. Recent years have witnessed growing concerns about the proliferation of harmful discourse online, including political polarization ([Barberá, 2015](#); [Garimella et al., 2021](#)), hate speech ([Poletto et al., 2021](#)), and cyberbullying. The SemEval–2026 Task 9 on Multilingual Polarization Detection formalizes this problem as a comprehensive framework for identifying and categorizing divisive content across languages and cultural contexts.

Current approaches to content moderation rely heavily on natural language processing (NLP) tech-

niques, with transformer-based models like BERT, ROBERTA ([Liu et al., 2019](#)), and DEBERTA demonstrating state-of-the-art performance. However, these models face significant challenges in multilingual settings, particularly for low-resource languages. While multilingual models such as XLM-R and cross-lingual transfer methods have made progress, significant performance gaps persist for African languages ([Nekoto et al., 2020](#)). This limitation is particularly problematic for Swahili (SW), which serves as a lingua franca for over 100 million speakers across East Africa but remains under-represented in current NLP resources ([Adelani et al., 2021](#)).

Let $\mathcal{X}$ denote the space of social media posts and $\mathcal{L} = \{\text{EN}, \text{SW}\}$ the language set. SemEval–2026 Task 9 decomposes polarization analysis into three supervised learning problems:

$$f_l^{(t)} : \mathcal{X} \rightarrow \mathcal{Y}^{(t)}, \quad t \in \{1, 2, 3\}, l \in \mathcal{L}$$

where:

- **Subtask 1:** $\mathcal{Y}^{(1)} = \{0, 1\}$ (binary polarization detection)
- **Subtask 2:** $\mathcal{Y}^{(2)} = \{0, 1\}^5$ (polarization type: political, racial/ethnic, religious, gender/sexual, other)
- **Subtask 3:** $\mathcal{Y}^{(3)} = \{0, 1\}^6$ (manifestation: stereotype, vilification, dehumanization, extreme language, lack of empathy, invalidation)

This formulation presents several technical challenges: (1) multilingual variation with limited annotated resources for Swahili, (2) severe label imbalance in multi-label tasks, (3) hierarchical label dependencies, and (4) cultural specificity of polarization expressions ([Ousidhoum et al., 2019](#)).

Our approach combines multiple strategies: multilingual adaptive fine-tuning, ensemble methods ([Dietterich, 2000](#)), weak supervision ([Ratner et al.,](#)

2017; Zhang et al., 2021), and class-aware optimization (Lin et al., 2017). Our contributions include:

1. A comprehensive framework for multilingual polarization detection across three subtasks
2. Theoretical analysis of ensemble variance reduction under correlated model errors
3. Practical implementation with language-specific ensembles and weak supervision
4. Extensive empirical evaluation on the official Codabench benchmark

2 Methodological Framework

2.1 Learning Framework

Given a dataset $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$, we minimize the regularized empirical risk:

$$\theta^* = \arg \min_{\theta \in \Theta} \left[\frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathbf{x}_i), \mathbf{y}_i) + \lambda \|\theta\|_2^2 \right] \quad (1)$$

where ℓ is a task-specific loss function and $\lambda > 0$ controls regularization.

2.2 Transformer Architecture

For transformer-based models, we use the standard architecture:

$$\mathbf{H} = \text{Transformer}(\mathbf{x}) = [h_1, h_2, \dots, h_T] \quad (2)$$

with classification head:

$$\hat{\mathbf{y}} = \sigma(\mathbf{W}h_{[\text{CLS}]} + \mathbf{b}) \quad (3)$$

where σ is the sigmoid activation for binary/multi-label classification.

3 Task-Specific Formulations

3.1 Subtask 1: Binary Polarization Detection

For binary classification $y \in \{0, 1\}$, we use binary cross-entropy loss:

$$\mathcal{L}_{\text{BCE}} = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})] \quad (4)$$

Proposition 3.1 (Gradient Stability). The gradient $\partial \mathcal{L}_{\text{BCE}} / \partial z = \hat{y} - y$ is bounded and ensures stable optimization.

3.2 Subtask 2: Multi-label Polarization Type Classification

For $C = 5$ polarization types with labels $\mathbf{y} \in \{0, 1\}^C$, we use independent binary classifiers:

$$\mathcal{L}_{\text{ML}} = - \sum_{j=1}^C [y_j \log \hat{y}_j + (1 - y_j) \log(1 - \hat{y}_j)] \quad (5)$$

3.3 Subtask 3: Multi-label Manifestation Identification

For extreme class imbalance, we define inverse frequency weights:

$$w_j = \min \left(C_{\max}, \frac{N}{\sum_{i=1}^N y_{ij}} \right) \quad (6)$$

with $C_{\max} = 100$ to prevent extreme values.

The weighted loss becomes:

$$\mathcal{L}_{\text{WBC}} = - \sum_{j=1}^C w_j [y_j \log \hat{y}_j + (1 - y_j) \log(1 - \hat{y}_j)] \quad (7)$$

3.4 Weak Supervision Framework

To address data scarcity, we implement keyword-based weak supervision:

$$\Phi(\mathbf{x}, \mathcal{K}_j^{(l)}) = \mathbb{I} \left(\sum_{k \in \mathcal{K}_j^{(l)}} \mathbb{I}(k \in \mathbf{x}) \geq \tau_j^{(l)} \right) \quad (8)$$

where $\mathcal{K}_j^{(l)}$ are language-specific keyword sets and $\tau_j^{(l)}$ are thresholds.

3.5 Ensemble Strategy

We employ language-specific ensembles:

$$\mathcal{E}_{\text{EN}} = \{\text{BERT-base, RoBERTa, DeBERTa}\}$$

$$\mathcal{E}_{\text{SW}} = \{\text{Afro-XLMR, XLM-R, BERT-multilingual}\}$$

Predictions are combined via probability averaging:

$$\hat{y}_j^{(\text{ensemble})} = \frac{1}{|\mathcal{E}_l|} \sum_{m \in \mathcal{E}_l} \sigma(z_j^{(m)}) \quad (9)$$

4 Theoretical Analysis

4.1 Ensemble Variance Reduction

Theorem 4.1 (Ensemble Variance Bound). Let $\mathcal{F} = \{f_1, \dots, f_M\}$ be an ensemble of models with

errors $\epsilon_m = f_m(\mathbf{x}) - y$. Define average pairwise correlation:

$$\bar{\rho} = \frac{2}{M(M-1)} \sum_{i < j} \rho(\epsilon_i, \epsilon_j)$$

Then the ensemble variance satisfies:

$$\text{Var}(\hat{y}_{\text{ens}}) \leq \frac{1 + (M-1)\bar{\rho}}{M} \max_m \sigma_m^2$$

where $\sigma_m^2 = \text{Var}(\epsilon_m)$.

Corollary 4.2. For independent errors ($\bar{\rho} = 0$), ensemble variance reduces by factor $1/M$.

4.2 Weak Supervision Error Analysis

Let $\mathcal{Y}_{\text{true}}$ be true labels and $\mathcal{Y}_{\text{weak}}$ be weak labels. The precision gap:

$$\Delta P_j = P_{\text{true}}^{(j)} - P_{\text{weak}}^{(j)} \quad (10)$$

quantifies the quality of weak supervision.

5 Experimental Setup

5.1 Hyperparameter Configuration

- Batch Size: 16
- Learning Rate: 2×10^{-5}
- Epochs: 3-10 (task-dependent)
- Max Sequence Length: 128-512 tokens
- Weight Decay: 0.01
- Warmup Ratio: 0.1
- Optimizer: AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Threshold: 0.5

5.2 Implementation Details

All models were implemented in PyTorch using the HuggingFace Transformers library. The code is publicly available at [GitHub](#). We use 5-fold cross-validation for robust evaluation and report mean and standard deviation across folds.

6 Results and Analysis

6.1 Subtask 1: Binary Polarization Detection

Table 2: Official Codabench performance for Subtask 1

Metric	English (EN)	Swahili (SW)	Average
Accuracy	0.7625 $\pm$ 0.015	0.8195 $\pm$ 0.012	0.7910
Precision	0.6909 $\pm$ 0.018	0.8101 $\pm$ 0.014	0.7505
Recall	0.6441 $\pm$ 0.020	0.8333 $\pm$ 0.013	0.7387
F1 (Binary)	0.6667 $\pm$ 0.019	0.8215 $\pm$ 0.013	0.7441
F1 (Macro)	0.7411 $\pm$ 0.016	0.8195 $\pm$ 0.012	0.7803
F1 (Micro)	0.7625 $\pm$ 0.015	0.8195 $\pm$ 0.012	0.7910

Table 1: Model selection for Subtask 1

Model	EN Acc.	SW Acc.	Avg F1	Params
DistilBERT-base	0.730	0.650	0.690	66M
BERT-base-uncased	0.760	0.550	0.655	110M
BERT-multilingual	0.740	0.800	0.770	110M
XLM-RoBERTa	0.763	0.820	0.780	125M

6.2 Subtask 2: Multi-label Polarization Type Classification

Table 3: Official Codabench performance for Subtask 2

Metric	EN	SW
F1 Micro	0.1862 $\pm$ 0.019	0.1220 $\pm$ 0.017
F1 Macro	0.1988 $\pm$ 0.020	0.1036 $\pm$ 0.015
Precision Micro	0.1429 $\pm$ 0.018	0.0917 $\pm$ 0.016
Precision Macro	0.2166 $\pm$ 0.022	0.1767 $\pm$ 0.020
Recall Micro	0.2674 $\pm$ 0.021	0.1823 $\pm$ 0.019
Recall Macro	0.3004 $\pm$ 0.025	0.3053 $\pm$ 0.025

Table 4: Per-label F1 scores for Subtask 2

Category	EN F1	SW F1
Political	0.2553 $\pm$ 0.035	0.1429 $\pm$ 0.028
Racial/Ethnic	0.4444 $\pm$ 0.045	0.0896 $\pm$ 0.025
Religious	0.2222 $\pm$ 0.038	0.0833 $\pm$ 0.024
Gender/Sexual	0.0000 $\pm$ 0.000	0.0588 $\pm$ 0.022
Other	0.0721 $\pm$ 0.027	0.1434 $\pm$ 0.029

6.3 Subtask 3: Multi-label Manifestation Identification

Table 5: Official Codabench performance for Subtask 3 ($N = 509$)

Language	F1 Macro	F1 Micro	Exact Match
English (EN)	0.2668 $\pm$ 0.022	0.2733 $\pm$ 0.021	0.0125 $\pm$ 0.005
Swahili (SW)	0.3527 $\pm$ 0.024	0.3612 $\pm$ 0.023	0.0172 $\pm$ 0.006
Average	0.310	0.317	0.015

Table 6: Per-manifestation F1 scores for Subtask 3

Manifestation	EN F1	SW F1
Stereotype	0.2364 $\pm$ 0.031	0.3960 $\pm$ 0.038
Vilification	0.3577 $\pm$ 0.041	0.4684 $\pm$ 0.043
Dehumanization	0.2474 $\pm$ 0.032	0.2091 $\pm$ 0.030
Extreme Language	0.3866 $\pm$ 0.042	0.3492 $\pm$ 0.040
Lack of Empathy	0.1200 $\pm$ 0.022	0.3453 $\pm$ 0.039
Invalidation	0.2526 $\pm$ 0.033	0.3481 $\pm$ 0.040

Table 7: Ensemble variance reduction across tasks

Subtask	Single Var	Ensemble Var	Red. (%)
Task 1 (EN)	0.0215	0.0098	54.4
Task 1 (SW)	0.0187	0.0079	57.8
Task 2 (EN)	0.0321	0.0152	52.6
Task 2 (SW)	0.0274	0.0127	53.6

Table 8: Precision comparison: weak supervision vs. human annotation

Category	Weak Sup.	Human Ann.
Political	0.22 ± 0.04	0.78 ± 0.03
Racial/Ethnic	0.18 ± 0.05	0.75 ± 0.04
Religious	0.16 ± 0.04	0.71 ± 0.05
Gender/Sexual	0.12 ± 0.06	0.68 ± 0.06
Other	0.15 ± 0.05	0.72 ± 0.04

6.4 Ensemble Effectiveness

7 Discussion

7.1 Cross-Language Performance Patterns

Our results reveal intriguing cross-language patterns:

$$\Delta_{\text{SW-EN}} = \begin{cases} +0.0784 & \text{Subtask 1 (Binary Detection)} \\ -0.0952 & \text{Subtask 2 (Type Classification)} \\ +0.0859 & \text{Subtask 3 (Manifestation Identification)} \end{cases}$$

This suggests Swahili models excel at detecting polarization and identifying manifestations, but struggle with categorizing polarization types compared to English models.

7.2 Weak Supervision Limitations

The failure on the "Gender/Sexual" category in Subtask 2 ($F1 = 0$) exposes limitations of keyword-based weak supervision:

7.3 Class Imbalance Challenges

The extreme class weights required (up to $w_{\max} = 279.3$) highlight severe data imbalance. This aligns with findings in Lin et al. (2017) regarding focal loss effectiveness for imbalanced scenarios.

8 Limitations and Future Work

8.1 Identified Limitations

1. **Synthetic Label Quality:** Precision gap up to 0.56 indicates poor keyword coverage.
2. **Extreme Class Imbalance:** Maximum weight $w_{\max} = 279.3$ indicates severe data scarcity.

3. **Language Performance Gap:** Inconsistent performance across languages and subtasks.

4. **Label Independence Assumption:** Binary relevance ignores label correlations.

5. **Cultural Nuances:** Current methods may not capture cultural specificity adequately.

8.2 Future Directions

1. Data Augmentation:

$$\mathcal{D}_{\text{aug}} = \mathcal{D} \cup \text{BT}(\mathcal{D}) \cup \text{EDA}(\mathcal{D}) \cup \mathcal{D}_{\text{human}}$$

2. **Model Architecture Improvements:** Label correlation modeling, per-class threshold optimization.

3. **Loss Function Refinement:** Focal loss for hard examples.

4. **Democratization Tools:** Leverage AutoNLU (Zhu et al., 2020) for accessible implementation.

5. **Cyberbullying Integration:** Incorporate insights from

9 Conclusion

We have presented a comprehensive framework for multilingual polarization detection across three SemEval-2026 subtasks. Our approach integrates theoretical analysis with practical implementation, achieving competitive results while providing insights into cross-lingual transfer, ensemble effectiveness, and weak supervision limitations. The system demonstrates particular strength in Swahili binary detection ($\text{Macro-F1} = 0.8195$), indicating promising directions for low-resource language NLP. Future work will focus on addressing data scarcity, improving weak supervision quality, and capturing cultural nuances in polarization expression.

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